Gargi and Raman hardware check

For the HPC convener / in-charges. About 10 minutes.

The HPC workload questionnaire compares what research groups report against the capacity of Gargi and Raman. These are the hardware and policy figures it currently uses. Where a figure is right, tick "Correct?". Where it is wrong or missing, type the right value in the last column. "Not known" is a fine answer. Approximate figures are welcome; please say so.

Fill this in a PDF reader (Adobe Acrobat Reader, macOS Preview, or a browser), save, and send it back.

Filled in by (name, role, date)

1. Gargi

Item	Value we currently use	Correct?	Correction / comment
Standard compute nodes in service Nodes users can actually run on today.	17 nodes	☐	
CPU and cores per standard node Cores available to jobs (after any reserved for the system).	2x AMD EPYC 9554, 128 cores	☐	
Memory per standard node	768 GB	☐	
High-memory nodes: count, cores, memory	3 nodes, 128 cores, 2304 GB	☐	
GPU nodes	None	☐	
Interconnect	200 Gbps Mellanox InfiniBand HDR	☐	
Parallel storage: system and usable size	DDN EXAScaler 200NVX2, 120 TB	☐	
Home : scratch split of that storage Or give the two sizes in TB.	50 : 50 (assumed)	☐	
Storage bandwidth	15 GB/s	☐	
Quotas: home per user / per group; scratch per user Write "none" if there is no quota.			
Scratch clean-up (purge) policy e.g. files older than 30 days deleted; or none.			
Local disk on compute nodes (size usable by jobs) e.g. /tmp on SSD, 400 GB; or none.			

2. Gargi queues

Item	Value we currently use	Correct?	Correction / comment
Apart from the high-memory and GPU queues, are any queues bound to particular nodes? Yes / No. We assume No: the other queues share all compute nodes. If Yes, say which nodes each queue can use.			
Queue: debug	1–2 node(s), 0.5 h, 4 job(s) at once per group; ceiling for all groups together: not known	☐	
Queue: short	1–3 node(s), 36 h, 3 job(s) at once per group; ceiling for all groups together: not known	☐	
Queue: medium	1–3 node(s), 72 h, 3 job(s) at once per group; ceiling for all groups together: not known	☐	

Item	Value we currently use	Correct?	Correction / comment
Queue: long	1 node(s), 120 h, 1 job(s) at once per group; ceiling for all groups together: not known	☐	
Queue: xlong	1 node(s), 168 h, 1 job(s) at once per group; ceiling for all groups together: not known	☐	
Queue: hmq (high-memory nodes)	1 node(s), 168 h, 1 job(s) at once per group; ceiling for all groups together: not known	☐	
Limits apply to	Each research group (faculty member and one group member), not each user	☐	
Do the per-queue job limits add up across queues? E.g. can a group run 3 jobs in short and 3 in medium at the same time? Yes / No / Not known.			
Jobs one group may have running at once, all queues together Or "none".	not known	☐	
Nodes one group's running jobs may occupy in total Or "none".	not known	☐	
Jobs one group may have waiting in the queue Or "none".	not known	☐	
Fair-share policy in use If yes, briefly: the share per group and the decay period, if known.	No	☐	
Can jobs from other queues run on the high-memory nodes when they are free? Yes / No / Not known.			

3. Raman

Item	Value we currently use	Correct?	Correction / comment
Standard compute nodes in service Nodes users can actually run on today (some older nodes may be down).	64 nodes	☐	
CPU and cores per standard node We assumed 2 sockets x 12 cores = 24. Please confirm.	2x Xeon Gold 6126, 24 cores	☐	
Memory per standard node	96 GB	☐	
High-memory node(s): count, CPU, cores, memory Cores assumed; please confirm the number of sockets.	1 node, Xeon Platinum 8160, 24 cores, 1152 GB	☐	
GPU nodes: count	1 node	☐	
GPUs per GPU node	not known	☐	
GPU model and memory per GPU e.g. NVIDIA V100 32 GB.	Volta generation (unconfirmed)	☐	
GPU node: CPU cores and memory	24 cores, 96 GB	☐	
GPU node: NVLink between GPUs? Yes / No / Not known.			
Interconnect	100 Gbps Intel Omni-Path + 1 Gbps Ethernet	☐	
Parallel storage: system and usable size	NetApp E5700, 104 TB	☐	

Item	Value we currently use	Correct?	Correction / comment
Home : scratch split of that storage Or give the two sizes in TB.	50 : 50 (assumed)	☐	
Storage bandwidth	6 GB/s	☐	
Quotas: home per user / per group; scratch per user Write "none" if there is no quota.			
Scratch clean-up (purge) policy			
Local disk on compute nodes (size usable by jobs)			
Expected remaining service life of Raman e.g. warranty end, planned retirement year, or not known.			

4. Raman queues

Item	Value we currently use	Correct?	Correction / comment
Apart from the high-memory and GPU queues, are any queues bound to particular nodes? Yes / No. We assume No: the other queues share all compute nodes. If Yes, say which nodes each queue can use.			
Queue: debug	1–4 node(s), 0.5 h, 4 job(s) at once per group; ceiling for all groups together: not known	☐	
Queue: short	5–8 node(s), 48 h, 8 job(s) at once per group; ceiling for all groups together: not known	☐	
Queue: medium	2–4 node(s), 72 h, 4 job(s) at once per group; ceiling for all groups together: not known	☐	
Queue: long	1 node(s), 120 h, 1 job(s) at once per group; ceiling for all groups together: not known	☐	
Queue: gpu (GPU node)	1 node(s), 24 h, 1 job(s) at once per group; ceiling for all groups together: not known	☐	
Queue: hmq (high-memory node)	1 node(s), 72 h, 1 job(s) at once per group; ceiling for all groups together: not known	☐	
Limits apply to	Each research group (faculty member and one group member), not each user	☐	
Do the per-queue job limits add up across queues? E.g. can a group run 3 jobs in short and 3 in medium at the same time? Yes / No / Not known.			
Jobs one group may have running at once, all queues together Or "none".	not known	☐	
Nodes one group's running jobs may occupy in total Or "none".	not known	☐	
Jobs one group may have waiting in the queue Or "none".	not known	☐	
Fair-share policy in use If yes, briefly: the share per group and the decay period, if known.	Yes	☐	

Item	Value we currently use	Correct?	Correction / comment
Can jobs from other queues run on the high-memory and GPU nodes when they are free? Yes / No / Not known.			

5. Both clusters

Item	Value we currently use	Correct?	Correction / comment
Usage charges	CPU INR 0.10/core-h (7-day queue 0.20); GPU INR 1.00/card-h	☐	
Planned downtime per year (days), each cluster Maintenance, power shutdowns, nodes out of service. Sets the committee's availability figure (0.95 assumed, about 18 days).			
Average utilisation over the last 12 months (%), each cluster From Slurm accounting, if available; otherwise not known.			
Can a Slurm accounting export be shared (per group, last 12 months)? Core-hours, GPU-hours, job sizes and waits per group. Used only to check reported figures against records.			
Anything else the committee should know			